MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS
FALL SEMESTER 2025
TEST 2

Last Name (Surname):

First Name (Given name):

UID:

Instructons:

(1) The question paper is in the **last page**. You may tear off the last page, but **return the questions paper** at the end of the test.
(2) Write answers in the attached A4 papers. Writings in draft papers will NOT be graded.
(3) Show your steps or reasonings clearly to receive full credits unless otherwise specified in the question.

Write your work in this page

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Test Paper (Fall 2025)

Writings on this page will not be graded

Name:

UID:

(1) (3 points) Let $f : A \to \mathbb{R}$ be a function. Define the meaning that the limit of f at a cluster point c of A.

(2) (a) (3 points) State the Mean Value Theorem (for derivatives).

(b) (5 points) Let $f : [0,1] \to \mathbb{R}$ be a differentiable function. Suppose $f'(x) = 0$ for all $x \in [0,1]$. Use the Mean Value Theorem to show that f is a constant function.

(3) (5 points) Let $I = [a,b]$; and let $f : I \to \mathbb{R}$ be a continuous function on I such that for each x in I, there exists y in I such that $|f(y)| \leq \frac{1}{2}|f(x)|$. Show taht there exists a point $c \in I$ such that $f(c) = 0$.

(4) Let $g : [0,1] \to \mathbb{R}$

$$g(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

(a) (6 points) Find all points of continuity of g. Explain your answer.

(b) (3 points) Determine if there exists a differentiable function $f : [0,1] \to \mathbb{R}$ such that

$$f'(x) = g(x).$$

for all $x \in [0,1]$. Explain your answer.